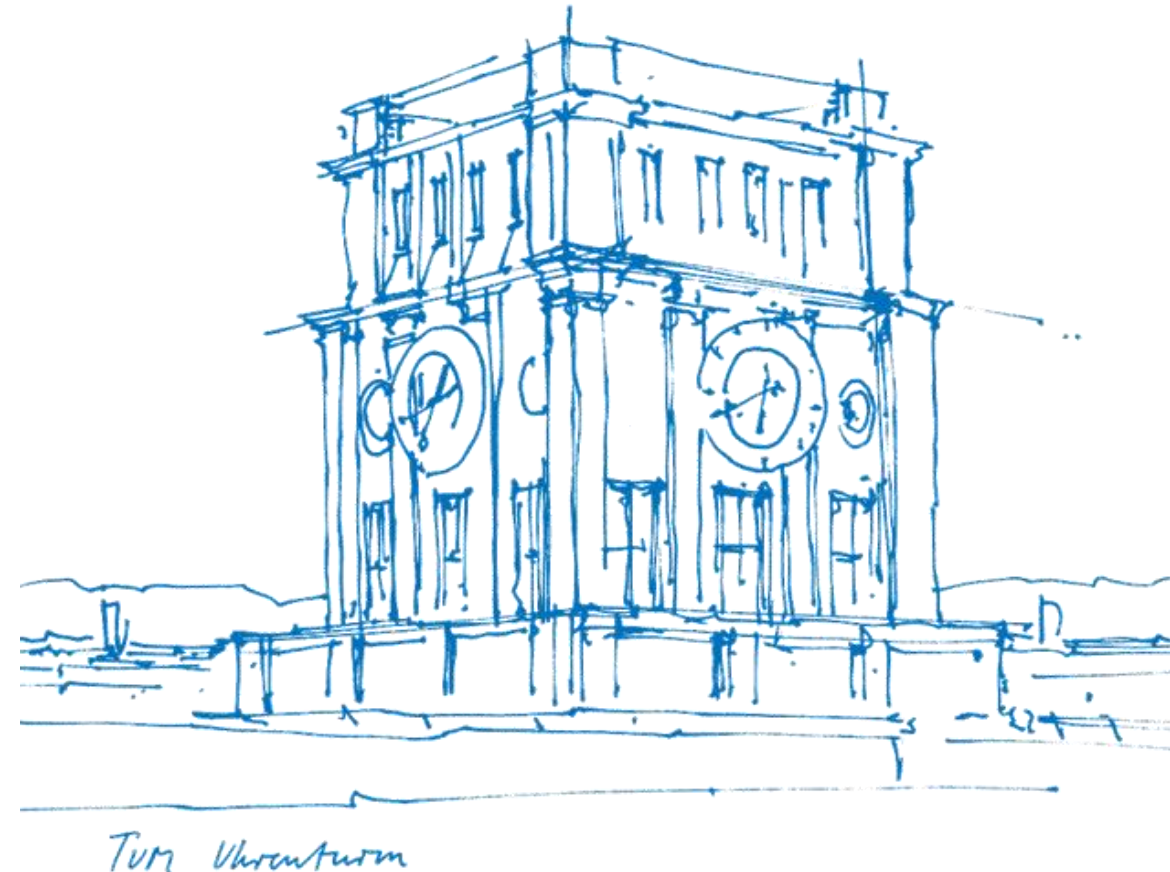


Tutorial 03

Measureable Map & Discrete Probability Measures

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About Tutorial

- **Tutorial:** In-Person, 2h by two time periods per week
 - 12:00-14:00 Tue, 03.11.018 (Group B33)
 - 10:00-12:00 Fri, 01.06.011 (Group E24)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Materials of this tutorial:** Please scan the QR-Code 

- **PART I.** Recap of the Lecture
- **PART II.** Exercises from the Exercise Sheet
- **PART III.** Additional Exercises (with solution)



I. Recap

- **σ -Algebra (Review)**
 - ① - Generator of σ -Algebra
 - ② - Induced σ -Algebra

- **Measurable map and random variable**
 - ③ - Measurable map
 - ④ - Random variable

- **Discrete probability measures**
 - ⑤ - Common Measures
 - Uniform distribution
 - Hypergeometric distribution
 - Bernoulli process
 - Binomial distribution
 - Geometric distribution
 - Poisson distribution

II. Exercise Sheet



Exercise 1. Let $(X, \mathcal{A}_X)$ and $(Y, \mathcal{A}_Y)$ be measurable spaces, $f : X \rightarrow Y$ a map, and $\mathcal{E}_Y \subseteq \mathcal{A}_Y$ arbitrary.

(i) Show that

$$\sigma(f^{-1}(\mathcal{E}_Y)) = f^{-1}(\sigma(\mathcal{E}_Y)).$$

(ii) Further, show that

$$f \text{ is } \mathcal{A}_X\text{-}\sigma(\mathcal{E}_Y)\text{-measurable} \iff f^{-1}(E') \in \mathcal{A}_X \text{ for all } E' \in \mathcal{E}_Y.$$

(iii) With that, finally show that for $\sigma(\mathcal{E}_Y) = \mathcal{A}_Y$ it holds that

$$f \text{ is } \mathcal{A}_X\text{-}\mathcal{A}_Y\text{-measurable} \iff f^{-1}(\mathcal{E}_Y) \subseteq \mathcal{A}_X.$$

This means that to show that a function is measurable, it is enough to show measurability on a generator.

Hint: From previous sheet 0 exercise 6, we have proved that the inverse image of a σ -algebra is still σ -algebra.

II. Exercise Sheet



Exercise 2. Let $\Omega \neq \emptyset$ be a set, $A \subseteq \Omega$, $A \neq \emptyset$, and let $\mathcal{E} \subseteq \mathcal{P}(\Omega)$. Then it holds that

$$\sigma(\mathcal{E}|_A) = \sigma(\mathcal{E})|_A.$$

Hint: From exercise 1, we have already proved that $\sigma(f^{-1}(\mathcal{E}_Y)) = f^{-1}(\sigma(\mathcal{E}_Y))$.

II. Exercise Sheet



Exercise 3. Let $(p_n)_{n \in \mathbb{N}} \subset [0, 1]$ with a fixed $\lambda := \lim_{n \rightarrow \infty} n p_n \in \mathbb{R}_{>0}$. Then for all $\omega \in \mathbb{N}$ we have that

$$\lim_{n \rightarrow \infty} p_{\text{Bin}(n, p_n)}(\omega) = p_{\text{Poi}(\lambda)}(\omega) .$$

Hint: Use the approximation formula $e^{-x} = \lim_{n \rightarrow \infty} \left(1 - \frac{x}{n}\right)^n$

II. Exercise Sheet



Exercise 4. After a long game night, Agathe and Balthasar are standing alone at the subway station Marienplatz, where a train departs every 10 minutes. Unfortunately, the trains are so overcrowded that only one person can board each time. Every person on the platform has the same chance of boarding the train. Additionally, until the next train arrives, two new people arrive at the station. Determine the probability that Agathe will be able to board

- (i) the n -th train,
- (ii) at the latest by the n -th train,

where $n \in \mathbb{N}$.

III. Additional Exercises

Exercise 5. Let X be a Poisson-distributed random variable with parameter $\lambda \geq 0$. Which equation does the probability mass function f_X satisfy for all $i \in \mathbb{N}_0$?

- ✓ $-f_X(i+1) = \frac{\lambda}{i+1} \cdot f_X(i)$
- $-f_X(i+1) = \frac{\lambda}{i} \cdot f_X(i)$
- $-f_X(i+1) = \frac{e^{-\lambda}\lambda}{i+1} \cdot f_X(i)$
- $-f_X(i+1) = \frac{e^{-\lambda}}{i+1} \cdot f_X(i)$

Poisson PMF Function: $f_X(i) = P(X=i) = \frac{e^{-\lambda}\lambda^i}{i!} \quad (i \in \mathbb{N}_0)$.

$$f_X(i+1) = \frac{e^{-\lambda}\lambda^{i+1}}{(i+1)!} = \frac{\lambda}{(i+1)} \cdot \frac{e^{-\lambda}\lambda^i}{i!} = \frac{\lambda}{(i+1)} f_X(i).$$

Let X be a Bernoulli-distributed random variable with success probability p . What is the probability-generating function of X ?

- $-G_X(s) = 1 + p + ps$
- ✓ $-G_X(s) = 1 - p + ps$
- $-G_X(s) = 1 - p + ps^2$
- $-G_X(s) = 1 - ps + ps^2$

Probability generating function (PGF) is defined as:
(Will learn in the future!)

$$G_X(s) = E(s^X) = s^0 \cdot P(X=0) + s^1 \cdot P(X=1) = (1-p) + ps = 1-p+ps.$$

Materials

